

Fig. 4: Charging capacitor output graph on Android

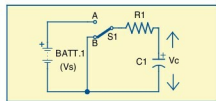


Fig. 5: Capacitor discharging circuit

and the right-side Y axis denotes the capacitor voltage. X axis indicates the charging time of the capacitor or time constant of the capacitor.

RC discharging circuit

In the discharging circuit, mechanical switch S1 is turned to position B and the battery is left open as shown in Fig. 5. During this time, the charge stored in the capacitor starts to discharge through resistor R1 until charge in the capacitor equals zero ($Q=0$).

From Fig. 6, at time $t=0$, V_c gradually decreases because accumulation of the positive charge and the negative charge freely moves away from the plates of the capacitor. So, the potential difference between the

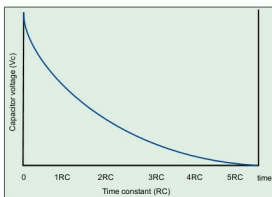


Fig. 6: Capacitor discharging voltage graph

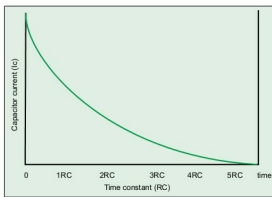


Fig. 7: Capacitor discharging current graph

plates decreases.

As shown in Fig. 7, from time $t=0$, current across the capacitor gradually decreases. In other words, at $t=0$, the battery is disconnected. So, stored electrostatic charges tend to freely move from the plates through the resistor. Therefore voltage (potential difference between two plates) across the two plates decreases. In the discharging circuit, both voltage and current gradually decrease.

Voltage across the capacitor and current in the capacitor are calculated using the formulae given in the box.

Time constant of the discharging capacitor is calculated using:

$$\tau = R \times C = R1 \times C1 = 50k \times 1000e^{-6} = 50 \text{ seconds}$$

For time $t=5\tau$ (five times constant), the capacitor is fully charged. So, we need to calculate 5τ to know the discharging time of the capacitor; $\text{time } t = 5(50k \times 1000\mu) = 250 \text{ seconds}$.

Capacitor maximum current $I_{\text{max}} = 0.003A$

Voltage across the discharging

capacitor is calculated using:

At time $t=0$, capacitor voltage = 15V; after 250 seconds, the capacitor has almost zero voltage:

$$V_c = V_s(e^{-t/RC}) \\ = 15(e^{-250/(50k \times 1000\mu)}) \\ = 0.101V$$

Current across the discharging capacitor $I_c = 2.0213\mu A$

At 54 seconds, the capacitor discharging voltage and capacitor discharging current are:

$$V_c = 5.093V \text{ and } I_c = 1.0187865e^{-4}$$

Voltage and current graphs of a discharging capacitor on Android are shown in Figs 8 and 9, respectively.

Development of Android application

First, you need to install Java Development Kit (JDK) version 1.7.0 or above on your Windows computer. We have tested this on Windows 8.1, 64-bit. Configuration of JDK is included in this month's EFY DVD accompanying EFY Plus magazine.

Next, you need to install Android Studio version 1.1.0. Details for setting up Android Studio, configuration of Software Development Kit (SDK) manager and Android Virtual Device have been included in the DVD.

Importing the Android project. To import the Android project, click file tab → Import as shown in the Fig. 10.

Click Android → Existing Android Code into Workspace as shown in Fig. 11.

Click Browse and navigate your Android project (rockt1). Click OK as shown in Fig. 12.

Program compilation. Click Android project folder; right-click on src folder → select run →. Click Run as Android application and select your phone from the dialog box or select the An-